

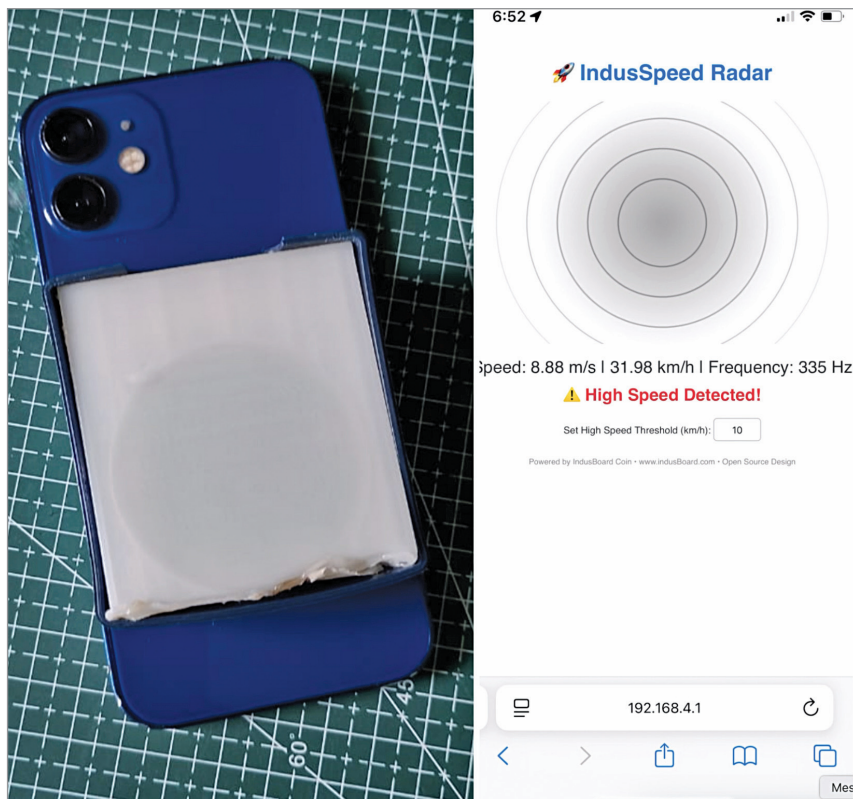


Fig. 9: Radar attached on a smartphone's back to show the real-time speed and Doppler frequency

frequency into speed is:

$$\text{Speed (m/s)} = \frac{\text{Doppler Frequency} \times \text{Wavelength}}{2}$$

For the HB100, the equation simplifies based on its known operating frequency and signal characteristics. The simplified empirical formula used in the code is:

$$\text{Speed (m/s)} = \text{pulse per second} \times 0.02652.$$

Here, 0.02652 is a constant derived from the wavelength of the HB100's microwave signal and its internal characteristics. This value provides an approximate conversion from pulse frequency to linear speed. To convert metres per second to kilometres per hour, the result is further multiplied by 3.6.

The calculated values are displayed via the serial monitor and also sent to a web page hosted by the IndusBoard Coin. The board creates a Wi-Fi access point (AP), allowing any smartphone or

computer to connect and view the real-time speed data.

A simple and interactive web dashboard displays the speed graphically like a radar plot, shows the current pulse frequency, and allows users to input a threshold speed limit. If the detected speed exceeds this threshold, a visual alert through a red colour is displayed on the web page.

The code can also be configured to operate in STA mode, allowing the system to work through a device

connected to an existing Wi-Fi network or mobile hotspot. Fig. 3 shows a snippet of the source code configuring Wi-Fi.

The source code should be first uploaded to the IndusBoard Coin by selecting the board as ESP32-S2 and choosing the appropriate port. The system can be then assembled in a suitable enclosure, whose case can be 3D-printed, if required. Finally, the radar, IndusBoard Coin, battery, and amplifier module should be positioned and secured as shown in Fig. 4 and Fig. 5.

Testing and deployment

After powering the system, it automatically creates a Wi-Fi hotspot named 'IndusRadar'. When a smartphone is connected to this hotspot, a browser can be opened and directed to the local IP address 192.168.4.1 to access the real-time speedometer dashboard.

Fig. 6 shows the radar measuring car speed on a road, alerting when high speed is crossed. Fig. 7 shows the web page UI displaying the radar waves and data in grey, frequency of Doppler effect, and speed in m/s and km/h.

Moving objects in front of the HB100 radar module, such as a car or a person walking, generate Doppler shifts, which the system converts to readable speed.

The threshold speed can also be adjusted using an input field, enabling customisation for various applications such as traffic speed enforcement or railway monitoring.

Fig. 8 shows the radar deployed on a roadside tripod to measure the speed of the incoming vehicles, while Fig. 9 shows the radar attached to a smartphone to show the vehicles' real-time speed and Doppler frequency. **EFY**

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